

Peter (Pedram) Beigi, PhD Candidate

Autonomous Driving Researcher and Machine Learning Engineer

Based in Santa Clara, CA & Washington, DC
Open to Relocate

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PROFESSIONAL SUMMARY

Develops perception, prediction, and decision-making systems for autonomous driving. Experienced in building scalable ML pipelines using large-scale multimodal datasets across simulation and real-world environments.

RESEARCH EXPERIENCE

Research Intern

May 2026 - Aug 2026

 University of Illinois Urbana-Champaign

Urbana-Champaign, IL

- Working with [Prof. Talebpour](#) on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.
- Developed a utility-guided residual multi-agent RL framework for decentralized spatial negotiation and trajectory generation in weakly structured traffic [\[paper\]](#) [\[code\]](#).
- Developed plan-conditioned diffusion world models for negotiation at unprotected conflict points [\[paper\]](#) [\[code\]](#).

Autonomous Vehicle & Machine Learning Intern

Sep 2022 - Present

 FHWA Turner-Fairbank Research Center (Saxton Lab)

McLean, VA

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools [\[code\]](#).
- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation [\[code\]](#).
- Built an end-to-end computer vision pipeline (Faster R-CNN, multi-camera MOT, BEV transformation, recursive state estimation, SIFT feature matching, and annotation/data processing) to extract trajectories and support motion prediction on 11,000+ vehicle-km of AV-human interaction data [\[code\]](#) [\[Data\]](#), **published by USDOT*.

Graduate Researcher

Sep 2021 - Present

 The George Washington University

Washington, DC

- Engineered model observability and traceability by developing Logsim DAGs to capture closed-loop metrics, improving experiment reproducibility for the autonomous stack.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

WORK EXPERIENCE

Autonomous Vehicle Researcher Intern

Sep 2024 - Dec 2024

  moment.ai (startup under **State Farm Insurance**)

Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

SELECTED PUBLICATIONS

h-index: 6 / citation: 246

"Decentralized Spatial Negotiation via Utility-guided Residual Multi-agent Reinforcement Learning for Trajectory Generation"	ICLR (under review)
"Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points" [pdf] [code]	IROS 2026
"Bridging Perception, Planning and Control: A System-Level Validation of a Microscopic Acceleration Model for Longitudinal Autonomous Driving" [pdf] [code]	IROS 2026
"Understanding the Interaction of Vehicles within an Intersection: An Optimal Control Approach with Differential Games"	INFORMS (2026) (Oral top 10%)
"Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control" [code]	IEEE VTC Fall (2026)
"Diffusion Process-Based Model for Network Trajectory Propagation" [pdf]	IEEE T-ITS (2026)
"Percolation-Based Cloud Computing Framework for Network Reliability Assessment" [pdf]	IEEE ICMI (2025)
"A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling" [code]	TRR (under review)
"A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance" [pdf] [code]	TRR (2025)
"Third Generation Simulation Dataset" [pdf] [code]	TRR (2024)

TECHNICAL SKILLS

Programming: Python, ROS, MySQL, Git

Libraries: PyTorch, Diffusers, Transformers, OpenCV, RLib, Gymnasium, Scikit-learn, Keras, Ultralytics

Cloud Computing Platform: Azure ML

Datasets: Waymo Motion, Waymo Perception, nuScenes, Argoverse 2, TGSIM, HighD, INTERACTION

Tools: CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

EDUCATION

The George Washington University

Sep 2021 - Exp: May 2027

PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

Sharif University of Technology

Sep 2016 - Feb 2021

Bachelor of Science in Civil Engineering - GPA: 3.80/4.00

RELEVANT COURSEWORK

- UMD: CMSC 422 Machine Learning (audited)
- UMD: CMSC 472 Deep Learning (audited)
- Stanford: CS231n DL for Computer Vision (Online)
- GW: CE 6609 Numerical Methods in Engineering (4/4)
- GW: CE 6800 Data Science (4/4)
- GW: ECE 6130 Big Data and Cloud Computing (4/4)
- UMD: CMSC 426 Computer Vision (audited)
- Google DeepMind: Introduction to RL (Online)

WORK AUTHORIZATION

No sponsorship required now or in the future.

Available to start immediately. Available with 3 days' notice for positions in the DC and Bay Area. Available with 2 week's notice for positions in other locations.